

# Demand System Asset Pricing: Theoretical Foundations

Fuchs, Fukuda, Neuhann

Discussant: Philippe van der Beck<sup>1</sup>

---

<sup>1</sup>Harvard Business School

# Background

- ▶ Given endowments, preferences, cash flow distribution, any asset pricing model implies
  1. an SDF that prices all assets:  $P = E[M \times \text{Cashflows}]$
  2. a demand system  $Q_i(P)$  such that  $\sum_i Q_i(P) = \text{Supply}$
- ▶ The demand-system approach is useful if portfolio holdings  $Q_i$  contain incremental information *beyond* prices  $P$ .
- ▶ When is that the case?
  1. Heterogenous portfolio holdings: Fact! ✓
  2. Demand elasticity  $< \infty$ : Strong evidence ( ✓)

# Background

- ▶ Downward-sloping demand curves featured in most asset pricing models.<sup>2</sup> (source: risk aversion)
  - ▶ But: Implied elasticity by standard models way too high!  
→ Falsely disregard quantity data
  - ▶ Very simple intuition: Unconstrained  $\Sigma^{-1}\mu$  implies crazy weights  
(unless we all agree, which seems off given the huge heterogeneity in choice sets in 13F data)  
→ Real portfolios don't look like this.
- ▶ Extensive (implied) evidence for low demand elasticities
  - ▶ **Equities:** Shleifer (1986), Harris and Gurel (1986), Wurgler and Zhuravskaya (2002), Greenwood (2008), Chang et al. (2014) etc.
  - ▶ **Fixed Income:** Greenwood and Vayanos (2014), Greenwood and Hanson (2013), Vayanos and Vila (2020) etc.
  - ▶ **FX:** Hau and Rey (2006), Gabaix and Maggiori (2015), Greenwood et al. (2019), Gourinchas et al. (2020)

---

<sup>2</sup> $Q = (\gamma\Sigma)^{-1}\mu \rightarrow \frac{\partial Q}{\partial P} = -(\gamma\Sigma)^{-1}$

# Background

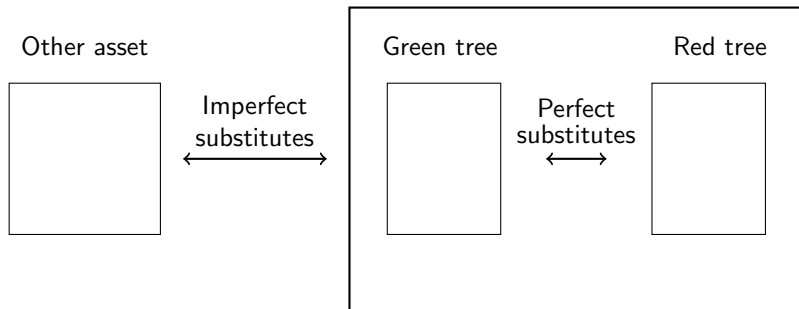
... Low demand elasticities are **not a “claim” from the recent literature** on demand-based asset pricing (neither a necessary condition)!

- ▶ DBAP jointly matches prices and investor-level data on portfolios, flows, trades.
- ▶ Justification: If markets are not infinitely elastic, then investor-specific demand shocks affect prices and portfolio holdings contain incremental information beyond observable market prices.

## Comment 1: One model versus another.

- ▶ Kojen and Yogo (2019) set up a model with certain assumptions (factor structure, myopic investors etc.)
- ▶ If you believe the assumptions (in particular the substitution patterns) you may trust the counterfactuals.
- ▶ But, the model has empirical support!
  1. Test 1: Model-implied elasticity on average very close to reduced form evidence from e.g. index inclusions  $\mathcal{O}(1)$ .
  2. Test 2: Assets with high latent demand have low expected returns.
- ▶ Note: KY (2019) not necessarily “taste-based”. Investor-specific loadings on characteristics could come from beliefs, risk aversion, constraints etc.
- ▶ You set up a different model with different substitution patterns.
- ▶ Naturally KY will not capture those.
- ▶ Can you show empirically how “bad” 1-factor structure (i.e. homogenous elasticities) assumption is?

## Comment 2a: “Substitution patterns are mismeasured”



- ▶ The model you set up features heterogenous substitution patterns:  $p_g$  and  $p_r$  always move in lockstep.
- ▶ Standard logit (KY 2019) estimated over the 2 + outside asset will (ironically) actually capture this!
- ▶ But 3 asset case + outside asset (with 2 different substitution elasticities) will not be captured by standard logit.
- ▶ Correctly identified **nested logit** framework will capture this. (Bretscher et al. 2023, Jiang et al. 2024, KY 2024)

## Comment 2b: “Elasticities have wrong magnitudes”

- ▶ **Experiment:** Increase in supply of green trees by  $\psi$ .
- ▶ **Equilibrium Theory:** Green and red tree are perfect substitutes.
- ▶ **Measured Elasticity:** Not perfect substitutes!
  - ▶ Price drops  $\Delta p_g / p_g = -\frac{\psi}{1+\psi}$ , and quantity increases  $\Delta a_g / a_g = 2\psi$
  - ▶ Your measured elasticity:  $-\frac{\Delta a_g / a_g}{\Delta p_g / p_g} = 2(1 + \psi) < \infty$ .

... This is not how elasticities are (ideally) measured in current DBAP!

- ▶ **Actual measured elasticity:** Indeed perfect substitutes!
  - ▶ What matters is **relative** prices and holdings of green vs red trees.
  - ▶ In equilibrium price of green tree and red tree both drop by  $-\frac{1}{1+\psi}$ !
  - ▶ The relative price (change) is  $\log p_g - \log p_r = 0$
  - ▶ Measured elasticity:  $-\frac{\log a_g - \log a_r}{\log p_g - \log p_r} = \infty$ .

## 2c: “even ideal instruments will suggest a low elasticity”

- ▶ Concern: Cannot use individual asset's (here green tree) instrumented price to estimate elasticity
- ▶ Demand systems are **not** identified from individually instrumented prices.
  - ▶ First stage takes entire cross-section into account and therefore includes spillovers
  - ▶ If price impact 0 in the cross-section (like for green vs red trees) then first stage irrelevant and second stage explodes (infinite elasticity).
- ▶ Formally
  1. Take exogenous demand shock in the cross-section  $\in \mathbb{R}^N$  (here  $[\psi, 0]$ )
  2. First stage: 
$$\begin{bmatrix} \Delta p_g \\ \Delta p_r \end{bmatrix} = c + \mathcal{M} \begin{bmatrix} -\psi \\ 0 \end{bmatrix} + \epsilon$$
    - reveals  $\mathcal{M} = 0$ .
    - Captures spillover effects!
    - Infinitely weak instrument.
  3. Second Stage Elasticity:  $-\frac{\text{cov}(\Delta q, [\psi, 0])}{\text{cov}(\Delta p, [\psi, 0])} = \infty$

## Other comments

... The paper points to many other shortcomings of the logit demand system. **All valid!** Very fruitful future research agenda. But do they invalidate existing applications?

- ▶ Comment 3: “**Dynamic trading (5.1)**”
  - ▶ See Gabaix and Koijen (2023) for dynamic demand system with forward looking expectations.
- ▶ Comment 4: “**Misspecified Constraints (5.3.1)**”
  - ▶ Short sales are not observed. May lead to mismeasured elasticities.
  - ▶ But empirically only for individual stocks, probably not for the entire cross-section.
- ▶ Comment 5: “**Endogenous capital flows (5.3.2)**”
  - ▶ Assumption of demand system: AUM distribution exogenous.
  - ▶ Get very similar estimates when this is endogenized (Azarmsa & Davis, 2023)

# Conclusion

- ▶ Demand systems are now widely used. Need better understanding on *how much/when* holdings data provide incremental information and when counterfactuals are reasonable.
- ▶ Useful dictionary between IO language and standard asset pricing.
- ▶ Valid point: heterogenous substitution patterns not well captured by standard logit. Future work should use empirical covariances among cashflows/returns to inform elasticity matrices and improve substitution patterns.

## Comments

1. Use same estimation as KY when comparing model-based and estimated elasticity → Cross-section in levels!
2. Perhaps framing is too negative.
3. Instead of “defeating” demand systems, better to improve them?  
(e.g. Chaudhary et al., 2023, make similar point but with positive framing)

# Concrete Suggestions

1. The main concern the paper raises is: Heterogenous substitution patterns are mismeasured.
  - ▶ Nested logit: Problematic because... (pre-specify groups etc.)  
Fosgerau, Monardo, and de Palma (2022), Monardo (2021)
  - ▶ How can we better estimate them? Especially in limiting case of perfect substitution, i.e. redundant assets.
  - ▶ For example: Instead of estimating elasticity matrix directly from holdings, plug in implied estimate from realized covariance matrix.
2. Pick your favorite counterfactual experiment in the literature: Show that the resulting counterfactual prices imply arbitrage opportunity.
  - ▶ In theory: Yes, redundant asset priced with homogenous elasticity leads to arbitrage.
  - ▶ Empirically, not sure. (Wurgler, Zhuravskaya 2002)

## Appendix

# Example: Demand Based Asset Pricing

- ▶ Simple Example with 2 investors
  1. Arbitrageurs' portfolio:  $Q_1 = \Sigma^{-1}\mu$
  2. Taste investors' portfolio:  $Q_2$
- ▶ Equilibrium expected returns are lower for assets with higher weight in  $Q_2$  than in market portfolio.
- ▶ Equilibrium prices are determined by
  1. 1-factor model:  $Q_1$  (don't need quantity data, standard asset pricing without holdings)
  2. 2-factor model: Market Portfolio and  $Q_2$  (uses quantity data, holdings-based asset pricing)